SOC-001

Multiple Failed Windows Logins

Splunk SOC Home Lab Incident Report

Classification	Controlled Lab Simulation
Severity	Medium
Status	Closed - Lab Exercise
Affected Host	PC01 / PC01.itec.local
Affected Account	soc-test
SIEM	Splunk Enterprise 10.6.0.5
Endpoint Telemetry	Windows Security Event Log + Sysmon
Incident Date	October 4, 2026

Purpose: Demonstrate an end-to-end SOC workflow: endpoint telemetry collection, authentication-event detection, SPL investigation, event correlation, detection engineering, and scheduled alerting.

Scope: All activity in this report was intentionally generated inside an owned home lab for defensive training. No external systems were targeted.

1. Executive Summary

A controlled authentication test was performed against the local Windows account **soc-test** on endpoint **PC01**. Windows Security telemetry was forwarded to Splunk Enterprise through the Splunk Universal Forwarder. Splunk identified repeated Windows Event ID **4625** failed logons. Analysis showed interactive logon type **2**, source IP **127.0.0.1**, status **0xC000006D**, and substatus **0xC000006A**, consistent with a valid account receiving an incorrect password.

A broader authentication timeline also contained Windows Event ID **4624** successful authentication activity for the same test account. The successful event observed in the captured timeline occurred before the later group of failed attempts, so the evidence does **not** establish that the repeated failures resulted in account compromise. The lab activity was mapped to MITRE ATT&CK; **T1110.001 - Password Guessing** because the simulation intentionally generated repeated password failures.

2. Environment and Data Flow

Component	Role in Investigation
PC01 (Windows 10 VM)	Monitored endpoint and source of authentication events

Sysmon	Additional endpoint telemetry for the broader SOC lab
Splunk Universal Forwarder	Forwarded Windows event data from PC01
Splunk Enterprise	Central SIEM used for search, analysis, correlation, and alerting
Windows Security Log	Primary evidence source for Event IDs 4625 and 4624

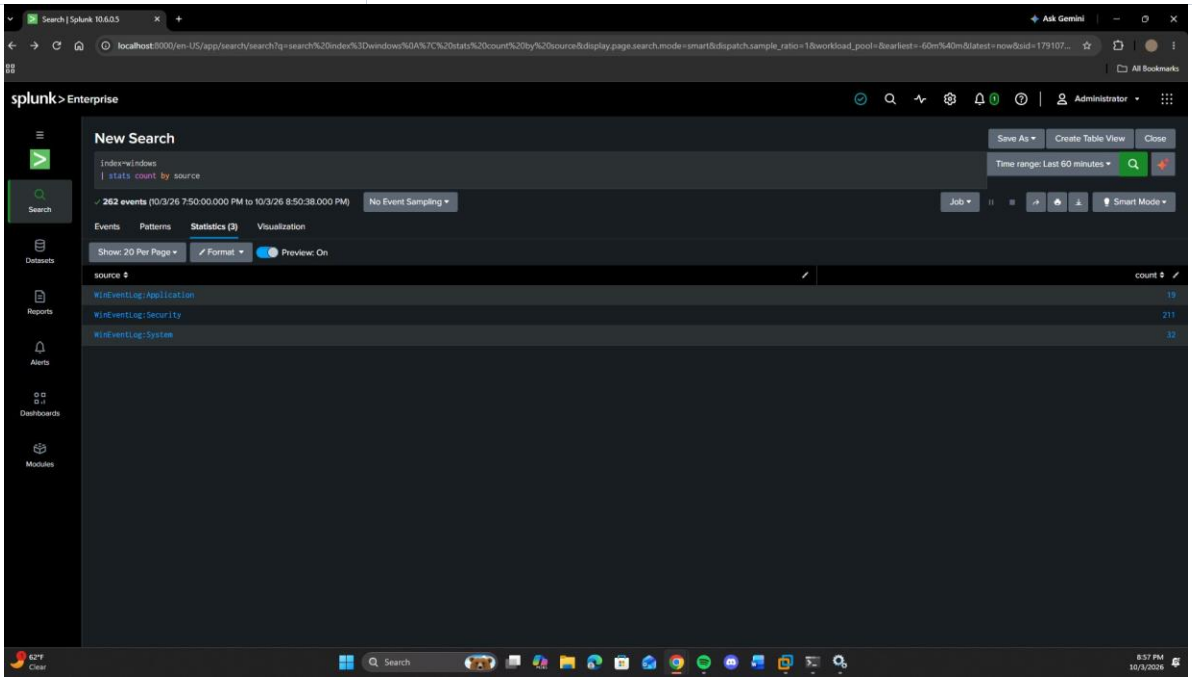


Figure 1. Splunk receiving Windows Application, Security, and System logs during pipeline validation. Sysmon forwarding was subsequently repaired after troubleshooting an event-channel permission issue.

3. Detection and Initial Evidence

The investigation began by searching the Windows Security source for Event ID 4625. Four failed authentication events were visible in the captured 15-minute window. The raw XML identified the target account as **soc-test** on **PC01**.

Initial SPL search

```
index=windows source="WinEventLog:Security" "4625"
```

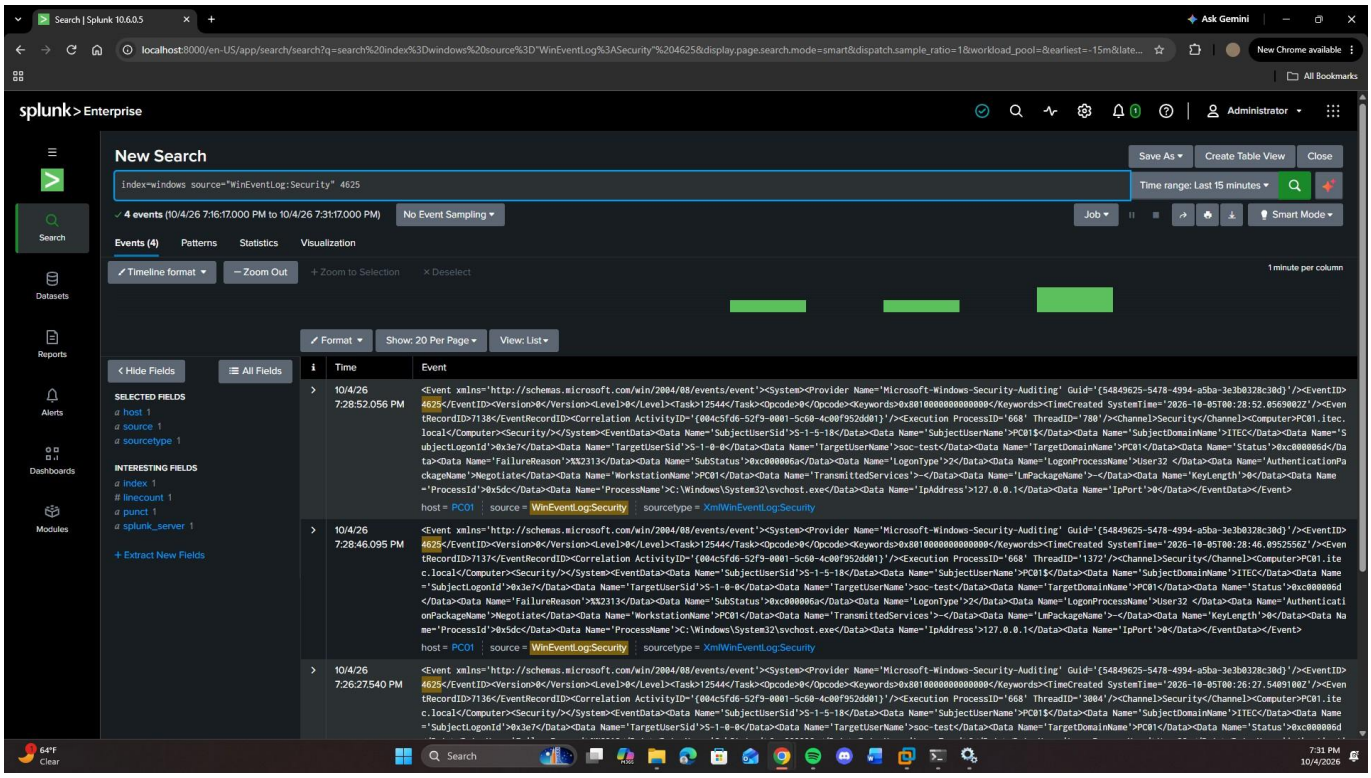


Figure 2. Raw Splunk results for Windows Event ID 4625 failed logons.

4. Event Analysis

Because the events were ingested as XML, SPL regular expressions were used to extract the fields needed for analyst review. The resulting table made the authentication evidence easier to interpret and document.

Key findings

Field	Observed Value	Analyst Interpretation
Target user	soc-test	Local test account targeted by the authentication attempts
Host	PC01	Windows endpoint that recorded the events
Event ID	4625	Windows failed-logon event
Logon type	2	Interactive logon
Source IP	127.0.0.1	Activity originated locally on the endpoint
Status	0xC000006D	Logon failure
SubStatus	0xC000006A	Incorrect password for an existing account

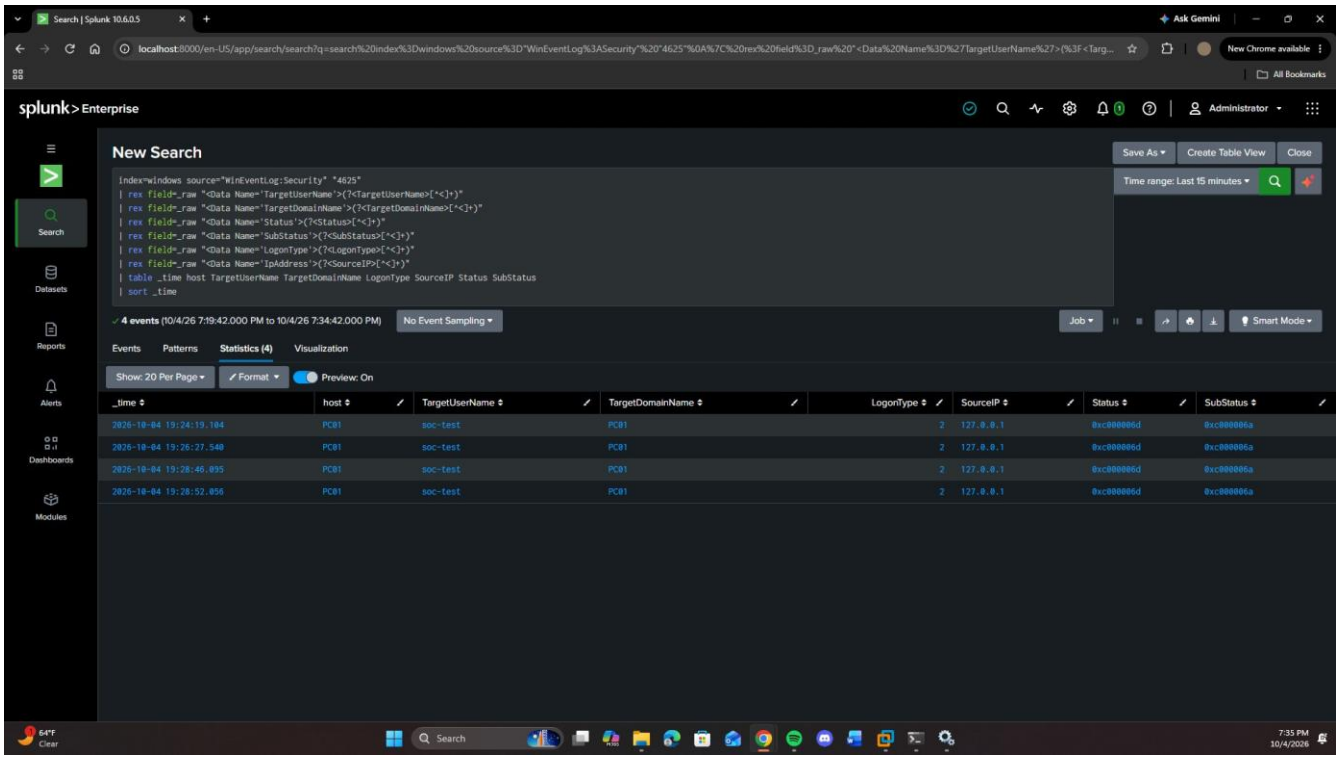


Figure 3. Extracted authentication fields used to analyze the failed logons.

5. Authentication Timeline and Correlation

The investigation correlated Windows Event IDs 4625 (failed logon) and 4624 (successful logon) for **soc-test**. The captured timeline contains repeated failures and one successful authentication during the broader search window. Importantly, the successful event at approximately 19:22 occurred before the later failures around 19:24-19:28. Therefore, this evidence should not be described as a successful brute-force compromise.

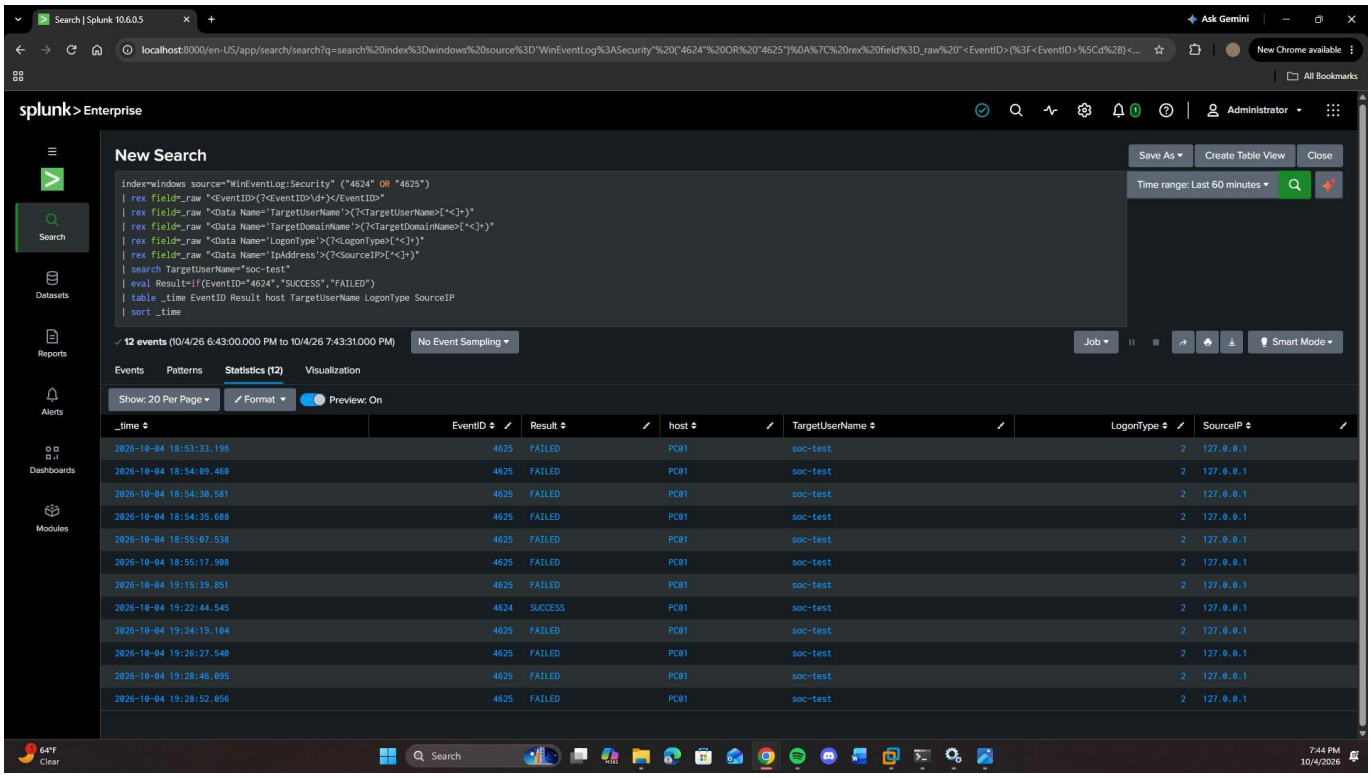


Figure 4. Authentication timeline correlating failed (4625) and successful (4624) events for soc-test.

Analyst assessment

The event pattern is suspicious enough to support a password-guessing detection in the controlled lab, but an analyst in a production environment would continue validating source context, user behavior, additional successful sessions, privilege changes, endpoint activity, and lateral movement before declaring compromise.

6. Detection Engineering

A threshold-based Splunk detection was created to identify accounts with five or more failed Windows authentication attempts during the search window. The detection groups activity by host, target user, source IP, and logon type, and records the first and last observed attempt.

Detection logic

```
index=windows source="WinEventLog:Security" "4625" | rex field=_raw "<Data
Name='TargetUserName'>(?<TargetUserName>[^<]+)" | rex field=_raw "<Data
Name='IpAddress'>(?<SourceIP>[^<]+)" | rex field=_raw "<Data Name='LogonType'>(?<LogonType>[^<]+)" |
stats count as FailedAttempts min(_time) as FirstAttempt max(_time) as LastAttempt by host
TargetUserName
 SourceIP LogonType | where FailedAttempts >= 5 | convert ctime(FirstAttempt) ctime(LastAttempt) | sort -
FailedAttempts
```

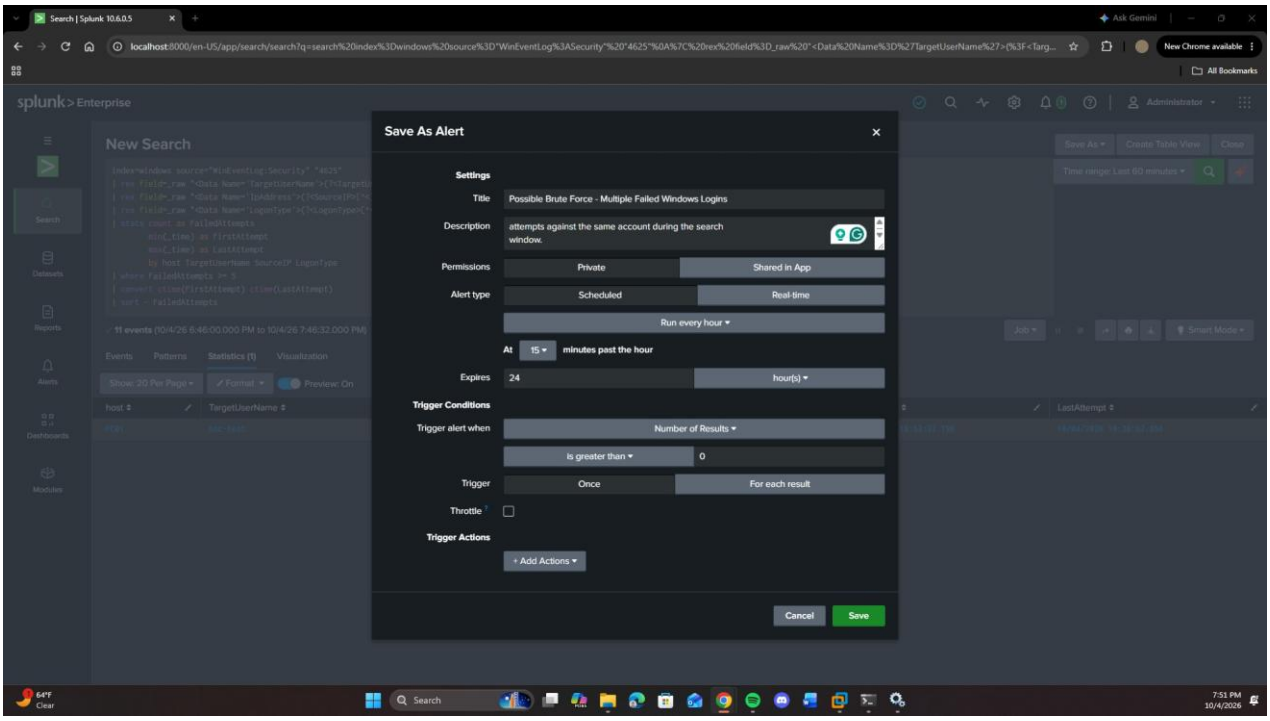


Figure 5. Splunk alert configuration for the repeated failed-login detection.

7. Alerting and Response

The detection was saved as an enabled scheduled Splunk alert named **Possible Brute Force - Multiple Failed Windows Logins**. The lab alert is intended to surface repeated authentication failures for analyst review rather than automatically block activity.

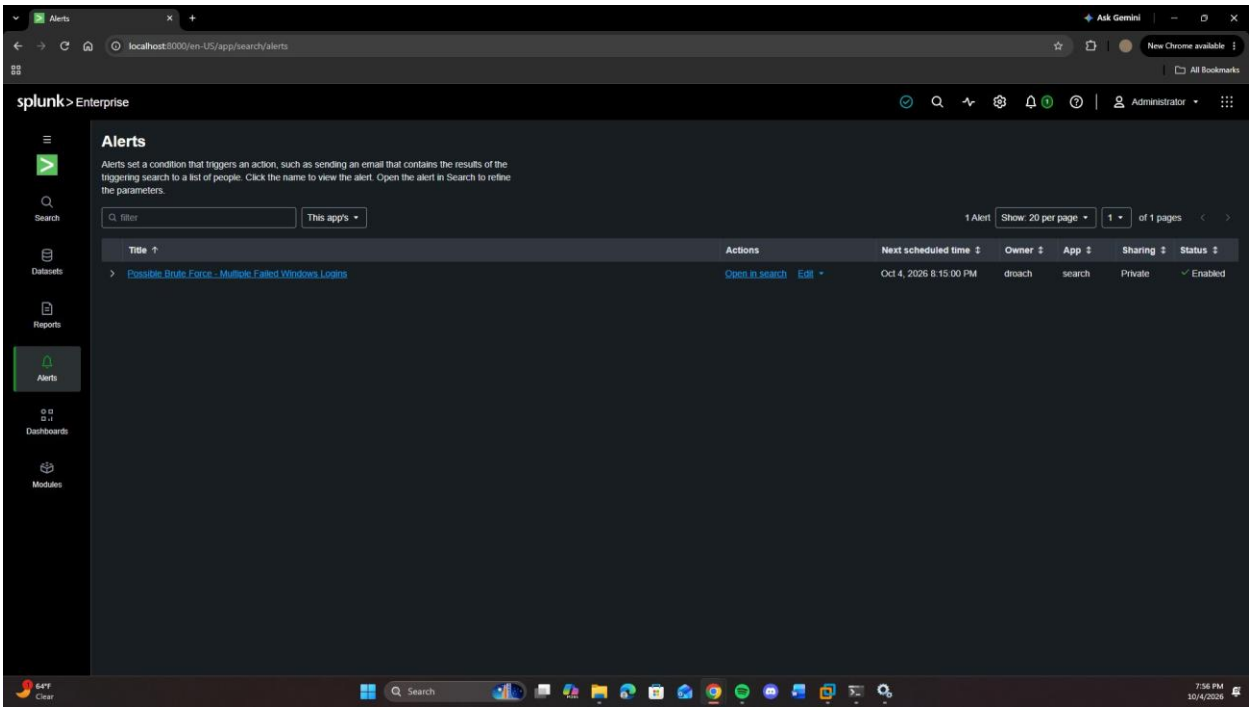


Figure 6. Saved and enabled Splunk alert.

Recommended response in a production environment

Validate: confirm the user, source system/IP, logon type, time window, and whether the activity is expected. **Correlate:** search for Event ID 4624 successes, privileged logons, account lockouts, endpoint process activity, and related network events. **Contain:** if compromise is suspected, disable or protect the account as appropriate, revoke active sessions, reset credentials, and block malicious sources where justified. **Scope:** hunt for the same source, username, or behavior across additional endpoints. **Recover:** restore normal access only after validating that credentials and affected systems are secure.

MITRE ATT&CK; mapping

T1110 - Brute Force

T1110.001 - Password Guessing

Mapping rationale: the controlled exercise intentionally generated repeated incorrect-password attempts against a known account. The mapping describes the simulated technique; the Windows logs alone do not prove a real external adversary was present.

8. Conclusion

SOC-001 successfully demonstrated an end-to-end defensive monitoring workflow. Windows authentication telemetry was collected from PC01, forwarded to Splunk, searched with SPL, normalized into analyst-friendly fields, correlated across failed and successful authentication events, and converted into a reusable scheduled detection. The investigation found repeated failed interactive logons against soc-test and documented the evidence without overstating the result as a confirmed compromise.

9. Lessons Learned

The lab reinforced several practical SOC skills: validating the telemetry pipeline before investigating an alert; reading raw Windows event XML; extracting fields when automatic parsing is insufficient; distinguishing evidence from assumptions; correlating multiple event IDs; and turning a one-time search into a repeatable detection. During setup, a Sysmon forwarding issue was traced to Windows event-channel access permissions (error code 5), providing additional troubleshooting experience with endpoint telemetry collection.

10. Evidence Inventory

Evidence	Purpose
Figure 1 - Splunk log sources	Validates Windows log ingestion
Figure 2 - Raw 4625 events	Shows original failed-logon evidence
Figure 3 - Extracted fields	Shows analyst-focused SPL parsing and review
Figure 4 - Authentication timeline	Correlates failed and successful authentication activity
Figure 5 - Alert configuration	Documents the detection-to-alert workflow
Figure 6 - Saved alert	Confirms the alert was enabled in Splunk

Portfolio note: This report documents a controlled home-lab exercise. Credentials and sensitive production data are not included.